Summative Assessment 1 | Section I: Task 3

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Setup

Each site contains the key phrase with probability p, independently. Let X be the number of sites searched until the first success, so $X \sim \text{Geometric}(p)$ with

$$P(X = k) = (1-p)^{k-1}p, \quad E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}.$$

```
set.seed(123)
p <- 0.6   # change this value to test other probabilities

# Simulation Function
# Number of sites searched until the key phrase is found
search_until_found <- function(p) {
  count <- 0
  found <- FALSE
  while (!found) {
    count <- count + 1
    # 1 = site contains the key phrase (prob p), 0 = it doesn't (prob 1-p)
    site <- sample(c(1, 0), size = 1, prob = c(p, 1 - p))
    if (site == 1) found <- TRUE
  }
  count
}

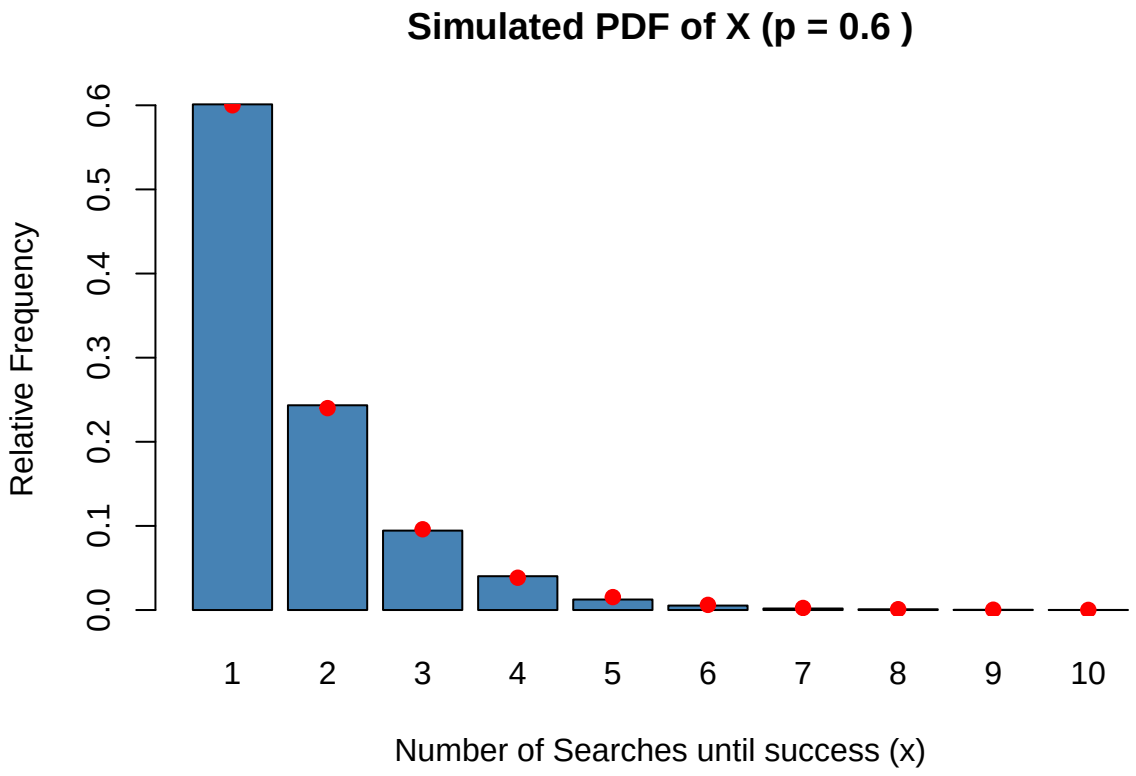
simulate_searches <- function(n = 10000, p = 0.6) {
  X <- numeric(n)
  for (i in 1:n) {
    X[i] <- search_until_found(p)
  }
  X
}
```

Part 1: Simulated PDF, Mean and Variance

```
n <- 10000
X <- simulate_searches(n, p)
```

```r
pdf_sim <- table(X) / n
bp <- barplot(pdf_sim, main = paste("Simulated PDF of X (p =", p, ")"),
              xlab = "Number of Searches until success (x)",
              ylab = "Relative Frequency", col = "steelblue")

# Theoretical PMF computed by: P(X = k) = (1 - p)^(k - 1) * p
k <- as.numeric(names(pdf_sim))
points(bp, (1 - p)^(k - 1) * p, pch = 19, col = "red")
```




```r
cat("Simulated Mean:    ", mean(X), "\n")
```

```
## Simulated Mean:     1.6497
```

```r
cat("Theoretical Mean:  ", 1 / p, "\n")
```

```
## Theoretical Mean:   1.666667
```

```r
cat("Simulated Variance:", var(X), "\n")
```

```
## Simulated Variance: 1.029493
```

```r
cat("Theoretical Variance:   ", (1 - p) / p^2, "\n")
```

```
## Theoretical Variance:    1.111111
```

The simulated pdf follows the theoretical geometric pmf closely, since the red points sit on top of the bars. The simulated mean is 1.65 against the theoretical $1/p = 1.667$, and the simulated variance is 1.029 against $(1-p)/p^2 = 1.111$. With 10000 runs the small gaps are just sampling error.

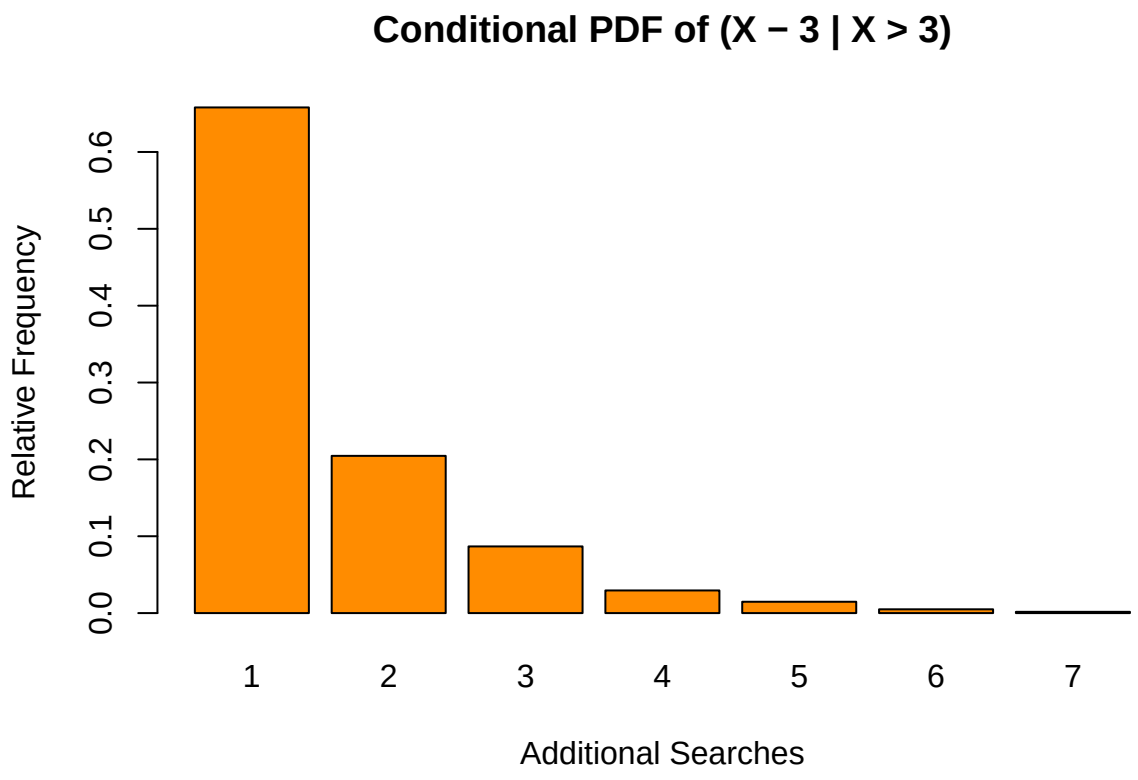
Part 2: Conditional Distribution given X > 3

```r
Xc <- X[X > 3]
Yc <- Xc - 3 # additional searches after the first 3 failures

cat("\nNumber of Runs with X > 3:", length(Xc), "\n")
```

```
##
## Number of Runs with X > 3: 611
```

```r
pdf_cond <- table(Yc) / length(Yc)
barplot(pdf_cond, main = "Conditional PDF of (X - 3 | X > 3)",
        xlab = "Additional Searches", ylab = "Relative Frequency",
        col = "darkorange")
```



```r
cat("Conditional Mean of X - 3:     ", mean(Yc), "\n")
```

```
## Conditional Mean of X - 3:      1.559738
```

```r
cat("Conditional Variance of X - 3: ", var(Yc), "\n")
```

```
## Conditional Variance of X - 3:  0.9222452
```

```r
cat("Compare with Full-Set Mean:  ", mean(X), " variance:", var(X), "\n")
```

```
## Compare with Full-Set Mean:   1.6497  variance: 1.029493
```

```r
# side by side check that the conditional pdf matches the full pdf
cmp <- rbind(full = as.numeric(pdf_sim[1:5]),
             conditional = as.numeric(pdf_cond[1:5]))
colnames(cmp) <- 1:5
print(round(cmp, 4))
```

```
##                  1      2      3      4      5
## full        0.6011 0.2434 0.0944 0.0402 0.0125
## conditional 0.6579 0.2046 0.0867 0.0295 0.0147
```

Only 611 of the 10000 runs had $X > 3$, so this conditional sample is small and noisier. The conditional pdf of $X - 3$ has a similar shape to the full pdf, though its mean (1.56) and variance (0.922) come out a bit lower than the full set values (1.65 and 1.029). With only 611 runs, a gap this size is plausible, and theory says the conditional distribution is exactly geometric with the same p, so the two are equivalent.

Part 3: Memoryless Property

```r
pa_cond <- sum(X == 4) / sum(X > 3)
pa_unc  <- mean(X == 1)
cat("\n(a) P(X=4 | X>3) =", pa_cond, "   P(X=1) =", pa_unc,
    "   Theoretical =", p, "\n")
```

```
##
## (a) P(X=4 | X>3) = 0.6579378    P(X=1) = 0.6011    Theoretical = 0.6
```

```r
pb_cond <- sum(X == 5) / sum(X > 3)
pb_unc  <- mean(X == 2)
cat("(b) P(X=5 | X>3) =", pb_cond, "   P(X=2) =", pb_unc,
    "   Theoretical =", (1 - p) * p, "\n")
```

```
## (b) P(X=5 | X>3) = 0.2045827    P(X=2) = 0.2434    Theoretical = 0.24
```

The conditional estimates are in the right range but not exact. $P(X = 4 \mid X > 3) \approx 0.658$ against $P(X = 1) \approx 0.601$, and $P(X = 5 \mid X > 3) \approx 0.205$ against $P(X = 2) \approx 0.243$.

Algebraically, $P(X > n) = (1-p)^n$, so

$$P(X = n + m \mid X > n) = \frac{(1-p)^{n+m-1}p}{(1-p)^n} = (1-p)^{m-1}p = P(X = m).$$